Latent Variable Moderated Mediation Using Correct and Misspecified Models

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Abstract

Moderated mediation is seldom estimated with latent variables, and its estimators have rarely been compared under model misspecification. In a preregistered simulation we evaluated three estimation approaches of a latent moderated mediation model, the local structural-after-measurement approach (LSAM) and two system-wide methods, across sample size, interaction magnitude, reliability, predictor distribution, and nine (mis)specification conditions. Under correct specification, all three recovered the interaction coefficient and index of moderated mediation, differing modestly in bias, mean squared error, coverage, standard errors, Type I error, and power. Under misspecification, LSAM was most robust, leaving some estimates unaffected by omissions that biased the system-wide estimators, although all misspecifications degraded performance.

Keywords: moderated mediation, structural equation modeling, latent interaction, measurement error, model misspecification

Latent Variable Moderated Mediation Using Correct and Misspecified Models

Mediation analysis is a well-known procedure in the behavioral and social sciences (Charlton et al., 2021; Hayes, 2018; Pieters, 2017; Preacher et al., 2007). In its basic and standard form, the effect of an independent variable on an outcome is decomposed into a direct effect and an indirect effect transmitted through a mediator, with all variables observed. This single mediator model has since been extended in a number of directions, most relevant here being the case in which the indirect effect is allowed to depend on a moderator (Edwards & Lambert, 2007; Hayes, 2015; MacKinnon et al., 2007; Montoya, 2024; Preacher et al., 2007).

In this context, a fundamental concern may arise from measurement. Many constructs under investigation are not directly observed but are measured indirectly through multiple items, that is, they are latent variables (Bollen & Hoyle, 2012). When such constructs are reduced to summary scores (e.g., a sum or mean of the items) and analyzed as though they were observed, a restrictive measurement structure is imposed rather than tested (McNeish & Wolf, 2020) and measurement error is left unmodelled. This error can attenuate or inflate the indirect effect and distort inferences about it (Cole & Preacher, 2014; Hoyle & Kenny, 1999), and it is especially consequential for the mediator, which acts at once as a predictor and an outcome (Gonzalez & MacKinnon, 2021). The problem is compounded under moderation, where the product term is even less reliable than either of its constituents (Bohrnstedt & Marwell, 1978; Busemeyer & Jones, 1983). To address this, structural equation modelling (SEM) with latent variables has been advocated, especially in the context of mediation (Cheung & Lau, 2008; Falk & Biesanz, 2015; Ledgerwood & Shrout, 2011; Montoya, 2024). It estimates structural relationships between common factors, from which indicator-specific unique variance has been partitioned, provided the measurement model is correctly specified (Bollen, 1989; DeShon, 1998; Rhemtulla et al., 2020). Yet we also note that latent variables do not necessarily improve power in general, whether for interactions or not (Cham et al., 2012; Fuller, 1987; Ledgerwood & Shrout, 2011; Wang & Rhemtulla, 2021).

Moderated mediation with latent variables is not straightforward in terms of statistical estimation. Indeed, a latent interaction[1] involves the product of two latent variables, which is unobserved and nonnormal even when its constituents are normal (Kelava & Brandt, 2022). Following this, a number of methods have been proposed to estimate nonlinear effects among latent variables (e.g., Algina & Moulder, 2001; Arminger & Muthén, 1998; Asparouhov & Muthén, 2021; Boker et al., 2023; Bollen, 1995; Jöreskog & Yan, 1996; Kenny & Judd, 1984; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Marsh et al., 2004; Mooijaart & Bentler, 2010; Mooijaart & Satorra, 2012; Wall & Amemiya, 2000). Most adopt a joint ("system-wide") estimation strategy, in which the measurement and structural parameters are estimated simultaneously. This is the case of the known methods: latent moderated structural equations (LMS) and product-indicator (PI). Opposed to these are structural-after-measurement (SAM) approaches (Cox & Kelcey, 2021; Holst & Budtz-Jørgensen, 2020; Ng & Chan, 2020; Rosseel et al., 2025; Wall & Amemiya, 2001, 2003), in which the measurement model is estimated first and the structural model afterward with the measurement parameters held fixed. Many of these approaches have so far been rarely applied in practice, likely held back by limited software, slow estimation, and error-prone implementations. With recent open-source implementations (Holst & Budtz-Jørgensen, 2020; Rosseel et al., 2025; Slupphaug et al., 2024) and non-commercial software (Keller, 2025), this is likely to change.

Beyond estimation, a further consideration concerns interpretation. Mediation is a hypothesis about causal effects, and causal interpretation requires identification assumptions that *may* not be verified from the data alone (Imai et al., 2010; Morelli et al., 2026; Rohrer et al., 2022; VanderWeele, 2015). With latent variable models, further caution is needed (Sterner et al., 2024; VanderWeele & Vansteelandt, 2022). We focus on one that is central to interpretation: the estimated latent variables must validly represent the intended constructs (Gonzalez & MacKinnon, 2021). In fact, when they do not, the model is subject to interpretational confounding

[1] We use *interaction* and *moderation* interchangeably, as the focus is on estimation, but we highlight they imply different interpretations and causal assumptions (VanderWeele, 2009, 2015). An interaction also need not be a product term, although that is the most common functional form used by researchers (Rimpler et al., 2026).

(Burt, 1976), a discrepancy between the empirical meaning assigned to a latent variable through its estimated measurement parameters and the meaning intended by the researcher. Under system-wide estimation, the consequences of a misspecification are not confined to the part of the model in which it occurs (Bainter & Bollen, 2014): the estimated loadings, and with them the empirical meaning of a latent variable, can change with the structural model that is imposed. Given that a SAM approach fixes the measurement model before any structure is fit, the estimated latent variables and their meaning do not depend on the structural model that follows, although this does not imply that the structure is correct (VanderWeele & Vansteelandt, 2022).

To illustrate interpretational confounding, we take a mediation model with three latent variables, X, M, and Y, each with three indicators, and obtain a population covariance matrix from a correctly specified version. We then alter it so that the covariances between the X and M indicators are unequal across the X indicators (3, 1, and 1, rather than the implied value of 2), so their association with M no longer follows a single common factor. To the same data we fit two structural models, one omitting the mediator and one including it, by system-wide SEM and by SAM (N = 200), and compare the estimated loadings of X. Without the mediator, both approaches recover the true loadings of one. Once the mediator enters, the system-wide loadings of x_2 and x_3 drop to 0.41, even though the measurement of X is unchanged, whereas the SAM loadings remain at one with and without the mediator.

Despite this, prior simulation work has examined latent interaction estimation mainly under distributional misspecification (Brandt et al., 2014; e.g., Cham et al., 2012; Lonati et al., 2024; Vieira et al., 2026), and in only one case under misspecification at the measurement level (Brandt et al., 2020), which considered only the UPI among the approaches studied here and in a limited setting of conditions. Moreover, when it comes to (moderated) mediation, despite its popularity in applied research, only a few studies have investigated it with latent variables, and those assumed a correctly specified (and saturated) model (Falk & Biesanz, 2015; Feng et al., 2020). In this article, we therefore evaluate three estimation approaches for a latent variable moderated mediation model: local SAM (LSAM; Rosseel et al. (2025)), LMS (Kelava et al., 2011; Klein & Moosbrugger, 2000), and the unconstrained PI approach (UPI; Kelava and Brandt (2009); Marsh et al. (2004)). These approaches have been explained extensively in previous research and we refer readers to it (e.g., Brandt et al., 2020; Kelava et al., 2011; Lin et al., 2010; Lonati et al., 2024; Rosseel et al., 2025; Slupphaug et al., 2024; Vieira et al., 2026).

Present Research and Contributions

This study makes two contributions. First, we extend Brandt et al. (2020), who examined misspecification only at the measurement level and, among the approaches considered here, only the UPI. We add LSAM and LMS and a considerably broader design, detailed in the next section, that includes misspecification of the structural model. Second, we evaluate the estimation approaches on the index of moderated mediation (imm; (Hayes, 2015)) itself, a nonlinear function of the structural parameters whose behavior need not follow that of its constituents. With latent variables, its performance has been examined only for system-wide estimators, under a correctly specified model, and with bias-corrected bootstrap inference (Cheung & Lau, 2017; Feng et al., 2020). Bootstrapping is computationally expensive, and its variants differ modestly in performance and rank differently depending on the criterion (Falk & Biesanz, 2015; Tibbe & Montoya, 2022). The Monte Carlo method (MacKinnon et al., 2004; Preacher & Selig, 2012) needs only the estimates and their covariance matrix, but has not been examined for the imm. We address that here.

Simulation Design and Performance Evaluation

In this section we describe the design of the simulation study and the measures used to evaluate performance, following some recent recommendations for the conduct and reporting of simulation studies (Morris et al., 2019; Pawel et al., 2025; Siepe et al., 2024).

Population Model and Estimands

We study a moderated mediation model among four latent variables: an exogenous predictor X, a moderator W, a mediator M, and an outcome Y. The structural model is as follows

$$M = a_1 X + a_2 W + a_3 (X \times W) + \zeta_M, \qquad Y = b\, M + c'\, X + \zeta_Y,$$

The primary estimands are the interaction coefficient (a_3) and the imm (i.e., a_3b). The remaining structural coefficients (a_1, a_2, b, c') and the twelve freely estimated factor loadings serve as secondary estimands. Note the imm is a derived quantity rather than a free parameter, hence its true value is computed as a_3b implied by the correctly specified model.

Population values were fixed at $a_1 = 0.4$, $a_2 = 0.3$, $b = 0.5$, and $c' = 0.15$, with $\text{cor}(X, W) = 0.3$. The interaction coefficient a_3 is a design factor (see below). These coefficients, together with the structural disturbance variances, were chosen so that under the baseline condition (normal exogenous predictors and a correctly specified model) each dependent latent variable has $R^2 = 0.30$ at the interaction level $a_3 = 0.20$. Also note that, because a_3 contributes to the variance of M, the explained variance is naturally higher with a larger a_3 and lower when its effect is null. Each latent variable is measured by four congeneric indicators with loadings 1.0, 0.8, 0.7, and 0.6. Figure 1 shows the population model, where the measurement and structural misspecifications studied below are drawn in grey.

Simulation Conditions

We employed a fully factorial design crossing five factors: sample size ($N \in \{100, 200, 300, 500\}$), interaction magnitude ($a_3 \in \{0, 0.20, 0.40\}$), measurement reliability (low = 0.5, high = 0.7), the distribution of the exogenous latent predictors (normal, uniform, t, and two χ^2 conditions with same- or opposite-direction skew), and the (mis)specification conditions (nine cases; see Data Generation below). This yielded $4 \times 3 \times 2 \times 5 \times 9 = 1{,}080$ conditions, each replicated 1000 times.

Sample sizes cover the small to moderate range, where interaction terms are estimated with limited power (Brandt et al., 2020; Lonati et al., 2024). The interaction magnitudes include a null condition to assess Type I error for the interaction coefficient and imm, and two nonzero levels representing small and moderate latent moderation effects. Reliability levels of 0.5 and 0.7 were chosen because measurement error is amplified in product terms and is therefore especially consequential for latent interaction estimation (Brandt et al., 2020; Lonati et al., 2024; Vieira et al., 2026). Nonnormality of the latent predictors is both empirically common and methodologically central for latent interaction estimators, whose distributional assumptions differ across approaches (Cham et al., 2012; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Lonati et al., 2024; Vieira et al., 2026). The misspecification conditions follow the moderated-mediation framework for the two structural omissions (Preacher et al., 2007) and prior latent interaction simulation work for the two measurement omissions (Brandt et al., 2020). They were also chosen as interpretable violations, acting respectively as a measured mediator–outcome confounder, a conflated moderated direct effect, exposure-correlated measurement error, and an unmeasured residual confounder (Edwards & Lambert, 2007; VanderWeele, 2015). The misspecification cases are detailed below.

Estimation Approaches

LSAM (Rosseel et al., 2025; Rosseel & Loh, 2024), with the measurement model estimated by ML in the first step. In this case, the interacting predictors X and W shared a single measurement block, while M and Y formed their own blocks. LMS (Klein & Moosbrugger, 2000), with robust standard errors, and the extended UPI approach with double mean centering (Kelava & Brandt, 2009; Lin et al., 2010; Marsh et al., 2004), referred to as UPI throughout, estimated with robust maximum likelihood (MLR), that is, ML with Huber–White sandwich standard errors.

Data Generation

Appendix A describes the complete data-generating mechanism in detail. Briefly, the latent variables X and W were drawn from a bivariate distribution with $\mathrm{Cor}(X, W) = \rho = 0.3$, constructed with the vine-to-anything (VITA) algorithm (Foldnes & Grønneberg, 2015; Grønneberg et al., 2022). Five exogenous distributions were examined, each standardized to mean 0 and variance 1. Thus, the conditions differ only in distributional shape, not in scale or location. Specifically, the normal condition draws $X, W \sim \mathcal{N}(0, 1)$ through a forced Gaussian copula, whereas the remaining four combine non-normal margins with a non-Gaussian copula: uniform, t_5, and two $\chi^2(2)$ conditions in which X and W are skewed in the same or in opposite directions. Given that, M and Y were generated from the structural equations above with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and $\zeta_Y \sim \mathcal{N}(0, \psi_Y)$, and the four indicators $(j = 1, \ldots, 4)$

of each latent variable $\eta \in \{X, W, M, Y\}$ were formed as $z_{\eta j} = \lambda_j \eta + \delta_{\eta j}$, with loadings $\lambda = (1, 0.8, 0.7, 0.6)$ and Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$ whose variances $\theta_{\eta j}$ were scaled to the target reliability $\omega \in \{0.5, 0.7\}$.

The nine (mis)specification conditions comprise one correctly specified condition and four types of misspecification, each at two empirically calibrated magnitudes (standardized 0.25 and 0.50, detailed in Appendix A; similar to Brandt et al. (2020)). Two are structural: an omitted direct effect of the moderator on the outcome (c_2) and an omitted moderated direct effect (c_3). The others are in the measurement model: a cross-loading of a mediator indicator on X (`cl`) and a residual covariance between a mediator and an outcome indicator (`rc`). In every case the effect is present in the population but omitted from the estimated model.

Nonconvergence, Outliers, and Inadmissible Solutions

Following previous research (Lonati et al., 2024; Vieira et al., 2026), replications were screened before analysis in two sequential steps, each applied globally so that every estimator is compared on an identical set of replications. First, we checked convergence. A fit was treated as nonconverged when any estimate, standard error, or confidence-interval bound of any structural coefficient or loading was missing or non-finite, and a replication was removed from all estimators whenever any one of them did not converge. Second, among the remaining replications, we screened for outliers, identified per condition, method, and parameter as point estimates beyond the interquartile range $\pm$ 3, together with replications whose estimated standard error exceeded 10 on any parameter of interest[2]. Again, a replication flagged as an outlier for any estimation approach was removed from all of them. We also flagged inadmissible solutions, where any residual (unique) variance was nonpositive or the covariance matrix of the exogenous latent variables and latent disturbances was not positive definite.

[2] We added the standard error criterion because some replications retained anomalous standard errors despite acceptable point estimates. It additionally removed 125 replications (about 0.01% of the globally converged ones), predominantly for UPI at the smallest sample size with low reliability and skewed χ^2 predictors.

Performance Measures

Performance was assessed with the measures implemented in the *simhelpers* package (Joshi & Pustejovsky, 2025), computed for the imm, the structural coefficients, and the loadings. These measures, with their empirical and Monte Carlo standard error (MCSE) calculations, are listed in Table 1. Note the MCSE are obtained in closed form where available and by jackknife otherwise. On top of that, where applicable, we also report the rejection rate (Type I error when $a_3 = 0$, power otherwise).

Inference for the imm followed the Monte Carlo method as mentioned before: for $r = 1, \ldots, R$ (with $R = 20{,}000$) we drew

$$\left(a_3^{(r)}, b^{(r)}\right) \sim \mathcal{N}_2\!\left((\hat{a}_3, \hat{b}),\, \hat{\Sigma}_{a_3 b}\right), \qquad \mathrm{imm}^{(r)} = a_3^{(r)}\, b^{(r)},$$

where $\hat{\Sigma}_{a_3 b}$ is the estimated covariance of $(\hat{a}_3, \hat{b})$, and took the 2.5th and 97.5th empirical percentiles of $\{\mathrm{imm}^{(r)}\}_{r=1}^{R}$ as the interval. The structural coefficients and loadings use Wald-based intervals from the model-implied parameter covariance matrix.

Reproducibility and Computational Details

All analyses were conducted in R (R Core Team, 2025) using the *lavaan* (Rosseel, 2012) and *modsem* (Slupphaug et al., 2024) packages for estimation, *covsim* (Grønneberg et al., 2022) and *rvinecopulib* (Nagler & Vatter, 2025) for data generation, and *simhelpers* (Joshi & Pustejovsky, 2025) for performance evaluation. Conditions were run in parallel with the *parallel* package. Reproducible and independent random-number streams across workers were obtained with the L'Ecuyer-CMRG generator. The computational environment was pinned with Nix (Dolstra et al., 2004) via the *rix* package (Rodrigues & Baumann, 2026) to ensure reproducibility, and the article was written with the *apaquarto* extension (Schneider, 2026). For the R version, package versions, code, manuscript, and materials, see the project repository.

Results

We present the results in three sections. First, the results for the interaction coefficient and the imm are reported. We then turn to the remaining structural coefficients (i.e., a_1, a_2, b, and c'), and we close with the estimated factor loadings, focusing mainly on that of the cross-loaded indicator (m_3). Moreover, due to the large number of conditions, we focused on a subset of the conditions. Namely, we fix the interaction coefficient at 0.2 in most cases (except for Type I error, which by construction requires a zero interaction) and display a limited number of sample sizes: $N = 200$ and 500 for the interaction coefficient and imm, and solely $N = 500$ for the structural coefficients and loadings. For the latter two, we further restrict to high reliability, whereas the interaction coefficient and imm are shown at both reliability levels. The entire set of results as well as detailed information about convergence and outliers is available in the repository (https://doi.org/10.5281/zenodo.20703258).

Convergence and Outliers

Convergence was high throughout. LSAM converged in every replication across all conditions, and LMS in at least 99.9%. UPI was the least stable, with its convergence rate falling to about 92% in the worst condition (the smallest sample size, low reliability, and skewed χ^2 predictors). At high reliability, however, UPI converged in at least 99% of replications regardless of sample size, distribution, or misspecification. Outliers were likewise rare, staying below 1% in the majority of conditions and reaching their highest values, up to roughly 11% for UPI and 8% for LSAM, only at $N = 100$ with low reliability and skewed (χ^2) predictors. Note the results reported below did not exclude inadmissible solutions. In fact, excluding them leads to negligible changes, with the few differences confined to the smallest sample size, low reliability, χ^2 conditions, mainly for UPI.

Main Structural Estimates

Relative Bias

We interpret values whose absolute relative bias exceeds 0.10 (10%) as biased. Under correct specification with normal predictors, all three estimators recover both the interaction coefficient and imm with no bias, shrinking even further as the sample size grows and at higher reliability. In fact, bias is negligible across most nonnormal predictors as well, with the exception of the χ^2 conditions, where LMS and LSAM are the most affected (Figure 2 and Figure 3). In this case, however, bias for LSAM is only present at low reliability and smallest two sample sizes.

Under misspecification, both c_2 and c_3 bias the imm for all three estimators, as expected. For the interaction coefficient, c_2 produces no bias, whereas c_3 leads to bias under the system-wide estimation approaches. Among the measurement omissions, `cl` attenuates the interaction coefficient only under the system-wide estimators, while for the imm LMS shows bias in the case of χ^2 conditions. Last, the `rc` leaves the interaction coefficient and imm essentially unchanged. In all cases the bias increases with the magnitude of the omission and at lower reliability (Figure 4). Also, in general, the extent of the bias is similar across the different distributions.

Note the bias of the interaction coefficient under c_3 is the one result not implied by the omission alone. As both omissions sit in the outcome equation, the mediator equation that defines the interaction coefficient stays correct. Hence, LSAM, which estimates the interaction coefficient from the mediator block alone, recovers it. The system-wide estimators fit the full covariance structure, where the latent interaction reaches the outcome only through the interaction coefficient and b, and absorb the omitted covariance into the interaction coefficient.

Relative RMSE

Under correct specification, relative RMSE falls with sample size and reliability for all three throughout distributions (Figure 5). Specifically, LMS is the most efficient, LSAM intermediate, and UPI the least. However, under misspecification, the structural omissions (c_2 and c_3) inflate the RMSE of the imm for all three estimators. In fact, such increase, in some cases, affects the system-wide estimators the most. With respect to the measurement omissions (`cl` and rc), we observed a negligible effect. For the interaction coefficient, only c_3 inflates slightly the RMSE mainly for UPI, while LSAM and LMS are unaffected. The inflation is larger at lower reliability.

Standard errors

We interpret SE/SD ratios below 0.90 (underestimation) or above 1.10 (overestimation) as biased, and similarly for the relative bias of the variance estimator. Note that the standard errors for the imm are based on the Monte Carlo method.

Throughout distributions and (mis)specifications, with high reliability, the results were very similar for both the interaction coefficient and imm. The SE/SD ratio is unbiased for all three approaches (Figure 6 and Figure 7). In case of low reliability and smaller sample size, we observed an overestimation for UPI throughout distributions. Moreover, also with low reliability and smaller sample size, in the χ^2 in opposite directions, LSAM and UPI struggled the most throughout all misspecifications, including with the correct one. With respect to the relative bias of the variance, apart from the similar observations as aforementioned, we further observe that some misspecifications affect in particular LMS in combination with specific nonnormal distributions (see Figure B1 and Figure B2 in Appendix B).

Coverage

Coverage between 92.5% and 97.5% was considered acceptable. Under correct specification with normal predictors, coverage is around the nominal level with all three approaches (Figure 8 and Figure 9) for the interaction coefficient and imm. Still with a correct specification, it goes down for both slightly under heavy-tailed (t_5) and χ^2 predictors for LMS and UPI, respectively, but only with a smaller sample size and low reliability.

Under misspecification, c_2 and c_3 reduce the imm coverage, since its interval is then centered on a biased estimate, with the largest drop for the system-wide estimators under c_3. This happens for both sample sizes, reliability levels, and most distributions, with the notable difference for the χ^2 in opposite directions, where it is less intense. For the interaction coefficient, the cross-loading (cl) is the most damaging case, collapsing coverage under the system-wide estimators across reliability levels, sample sizes, and distributions, while LSAM holds near nominal. The remaining omissions leave coverage close to nominal.

Hypothesis Testing (Type I error and Power)

Under correct specification with normal predictors and $a_3 = 0$, the interaction coefficient and imm holds its nominal level reasonably well for LSAM and LMS, while UPI is slightly conservative in the smaller sample size (Figure 10). Under misspecification still with normal predictors, only c_3 inflates the false-positive rate of the system-wide estimators. However, with nonnormal predictors, LMS shows inflated Type I error across multiple (mis)specifications.

At $a_3 = 0.2$ the imm is detected with power that rises with sample size and reliability (Figure 11), with LMS always displaying the highest power. Moreover, power falls substantially under opposite-skew χ^2 predictors, and most for UPI and LSAM. We note, however, that interpreting power in the case of bias and inflated Type I error should be avoided (Morris et al., 2019).

Additional Structural Estimates

For the remaining structural coefficients (a_1, a_2, b, and c'), at high reliability and $N = 500$, we report their relative bias (Figure 12), coverage (Figure 13), and SE/SD ratio (Figure 14) across all misspecification conditions and distributions.

Under correct specification, all approaches produced unbiased results, acceptable coverage and SE/SD ratios throughout distributions. With misspecifications, however, for the relative bias, the two structural omissions act on the outcome equation: c_2 biases the mediator-outcome slope b and, more strongly, the direct effect c', and c_3 is absorbed mainly by b and c' as well. Under LSAM the mediator-equation coefficients a_1 and a_2 stay essentially unbiased under both structural omissions, since LSAM estimates the mediator block in isolation. The system-wide estimators, by contrast, propagate c_2 into the mediator equation, biasing a_2 and likewise carry part of c_3 into c'. The cross-loading (`cl`) is the most disruptive, biasing all four coefficients for every estimator, because the omitted X loading on m_3 distorts the mediator's relations to the other latent variables. The residual covariance (`rc`) slightly inflates b and biases c' downward for all three. Across conditions the bias is consistently smaller under LSAM than under the system-wide estimators.

Moreover, coverage follows this bias pattern: near nominal where bias is small and dropping where bias is large, most sharply under the cross-loading and, for the outcome-side coefficients b and c', under c_2 and c_3. The SE/SD ratio stays close to one for all three estimators, with one clear exception: for b and c' under c_3 the standard errors are underestimated, most pronounced under heavy-tailed t_5 predictors.

Measurement Loadings

For the twelve freely estimated factor loadings, all but one are recovered with near-nominal performance across every measure and condition. Hence, we display only the cross-loaded mediator indicator m_3, with its relative bias, coverage, SE/SD ratio, and relative bias of the variance estimator shown together at high reliability and $N = 500$ (Figure 15). For m_3, the omitted cross-loading (`cl`) is the only consequential misspecification: its relative bias rises sharply (above 0.5 at the medium level, somewhat less for LSAM) and its coverage collapses toward zero. The two standard error measures, by contrast, yielded, to a certain extent, acceptable deviations: across conditions the SE/SD ratio and the relative bias of the variance estimator both stay within the desirable range, deviating strongly under the largest cross-loading, and the most so for LSAM. We also highlight that a subtler pattern, negligible in size, appears under the residual covariance (`rc`): the system-wide estimators carry a small systematic upward bias in m_3, under one percent, whereas LSAM stays centered on zero.

Discussion

We evaluated three estimation approaches in the context of a latent variable moderated mediation model, the two-step LSAM and the system-wide LMS and UPI, under correct specification and under a large number of structural and measurement misspecifications as well as distributional violations.

Under correct specification, all three recovered the imm and the interaction coefficient with negligible relative bias and near-nominal coverage, differing mainly in terms of coverage, Type I error, and the SE/SD ratio on one side, and relative RMSE and power on the other. LMS had the lowest relative RMSE and the highest power but showed below-nominal coverage and inflated Type I error, increasingly so under nonnormal predictors. LSAM yielded coverage and Type I error closest to nominal, at the cost of a higher relative RMSE and lower power in some cases, while UPI fell in between, accurate in relative bias but the least stable at the smallest sample size and low reliability. The two standard error performance measures, the SE/SD ratio and the relative bias of the variance estimator, showed similar results: both stayed near one at high reliability but not at low reliability and small samples. Indeed, UPI overestimated standard errors, and UPI and LSAM struggled the most under the χ^2 predictors skewed in opposite directions.

Under misspecification, LSAM was the more robust approach. The bias from an omission was more limited in LSAM than under the system-wide estimators: the structural omissions in the outcome equation (c_2, c_3) biased the outcome-side coefficients but left the mediator equation, and hence the interaction coefficient, unaffected, whereas LMS and UPI also biased the interaction coefficient. The omitted cross-loading (`cl`) was the most damaging throughout, biasing every coefficient and sharply lowering the coverage of the interaction coefficient under the system-wide estimators at low reliability, while LSAM remained near nominal; the residual covariance (`rc`) had little effect. Across conditions the relative bias was consistently smaller under LSAM, and it increased with the magnitude of the omission and at lower reliability.

These patterns are broadly consistent with previous research. The results for the LMS and UPI approaches under specific types of nonnormality alongside its stability under normality reproduces earlier findings (Brandt et al., 2014; Cham et al., 2012; Lonati et al., 2024; Vieira et al., 2026) and the behavior of LSAM also mirrors previous studies (Rosseel et al., 2025; Vieira et al., 2026). Moreover, part of measurement misspecifications results is aligned with Brandt et al. (2020). There, omitting a cross-loading was likewise a damaging misspecification, and the UPI approach was biased under it, whereas the two-stage method of moments on which LSAM builds attained the smallest RMSE in those conditions. Even so, that study found no estimator free of bias and none uniformly best, as we also observed.

Finally, although this article evaluated the estimation approaches from a statistical, inferential perspective, the moderated mediation model itself rests on strong causal assumptions (VanderWeele, 2015). This carries a practical consequence for applied researchers: good statistical performance of the estimation approaches under investigation in certain conditions is not, on its own, a justification for a causal research question or claim. Work addressing the causal interpretation of mediation continues to develop, and we direct interested readers to this literature (Imai et al., 2010; Morelli et al., 2026; Rohrer et al., 2022). Moreover, as we showed, misspecification has strong consequences across all three estimation approaches, and many of those may be highly plausible in certain context and fields.

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Table 1

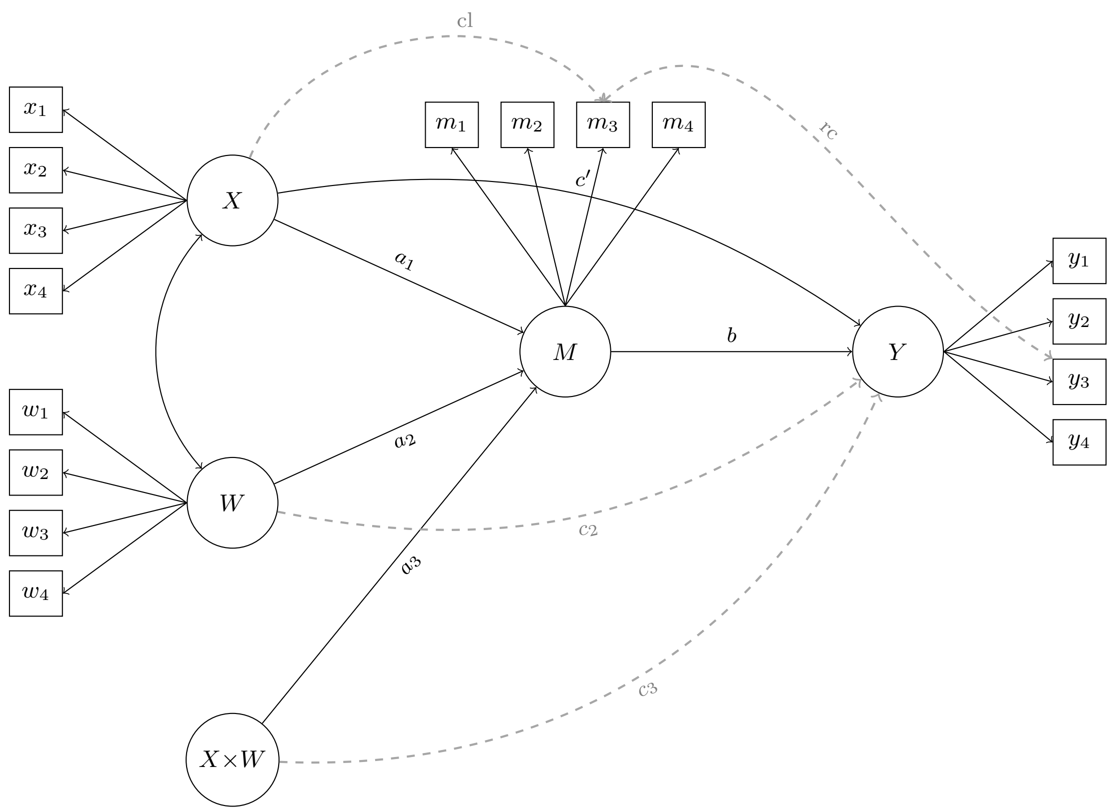
Subset of the Performance Measures for Evaluating Parameter Estimates

Performance measure	Empirical calculation	MCSE
Relative criteria		
Relative bias	$\frac{\bar{\hat{\theta}}}{\theta} - 1$	$\sqrt{S^2_{\hat{\theta}}/(K\theta^2)}$
Relative RMSE	$\frac{1}{\lvert\theta\rvert}\sqrt{\frac{1}{K}\sum_{k=1}^{K}(\hat{\theta}_k - \theta)^2}$	$\frac{1}{\lvert\theta\rvert}\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\text{RMSE}_{(j)} - \text{RMSE})^2}$
Standard errors		
Mean SE	$\overline{SE} = \frac{1}{K}\sum_{k=1}^{K}\widehat{SE}_k$	—
SE/SD ratio	$\overline{SE}/S_{\hat{\theta}}$	—
Relative bias of variance estimator	$\rho = \overline{SE^2}/S^2_{\hat{\theta}}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\rho_{(j)} - \rho)^2}$
Coverage		
Coverage	$c = \frac{1}{K}\sum_{k=1}^{K} I(L_k \le \theta \le U_k)$	$\sqrt{c(1-c)/K}$
Hypothesis testing		
Rejection rate	$r = \frac{1}{K}\sum_{k=1}^{K} I(0 \notin [L_k, U_k])$	$\sqrt{r(1-r)/K}$

Note. $\hat{\theta}_k$ is the estimate of parameter θ in replication k ($k = 1, \ldots, K$ usable replications), $\bar{\hat{\theta}}$ its mean, $S^2_{\hat{\theta}}$ its sample variance, and $S_{\hat{\theta}}$ its standard deviation; RMSE $= \sqrt{\frac{1}{K}\sum_k(\hat{\theta}_k - \theta)^2}$; $\widehat{SE}_k$ is the estimated standard error, $\overline{SE^2} = \frac{1}{K}\sum_k \widehat{SE}^2_k$, and L_k, U_k the confidence-interval limits; c and r denote coverage and rejection rate, and ρ the relative bias of the variance estimator. The subscript (j) marks a leave-one-out jackknife replicate. The MCSEs for relative RMSE and the relative bias of the variance estimator are computed by jackknife; the others are closed-form, and dashes mark MCSEs that are not computed.

Figure 1

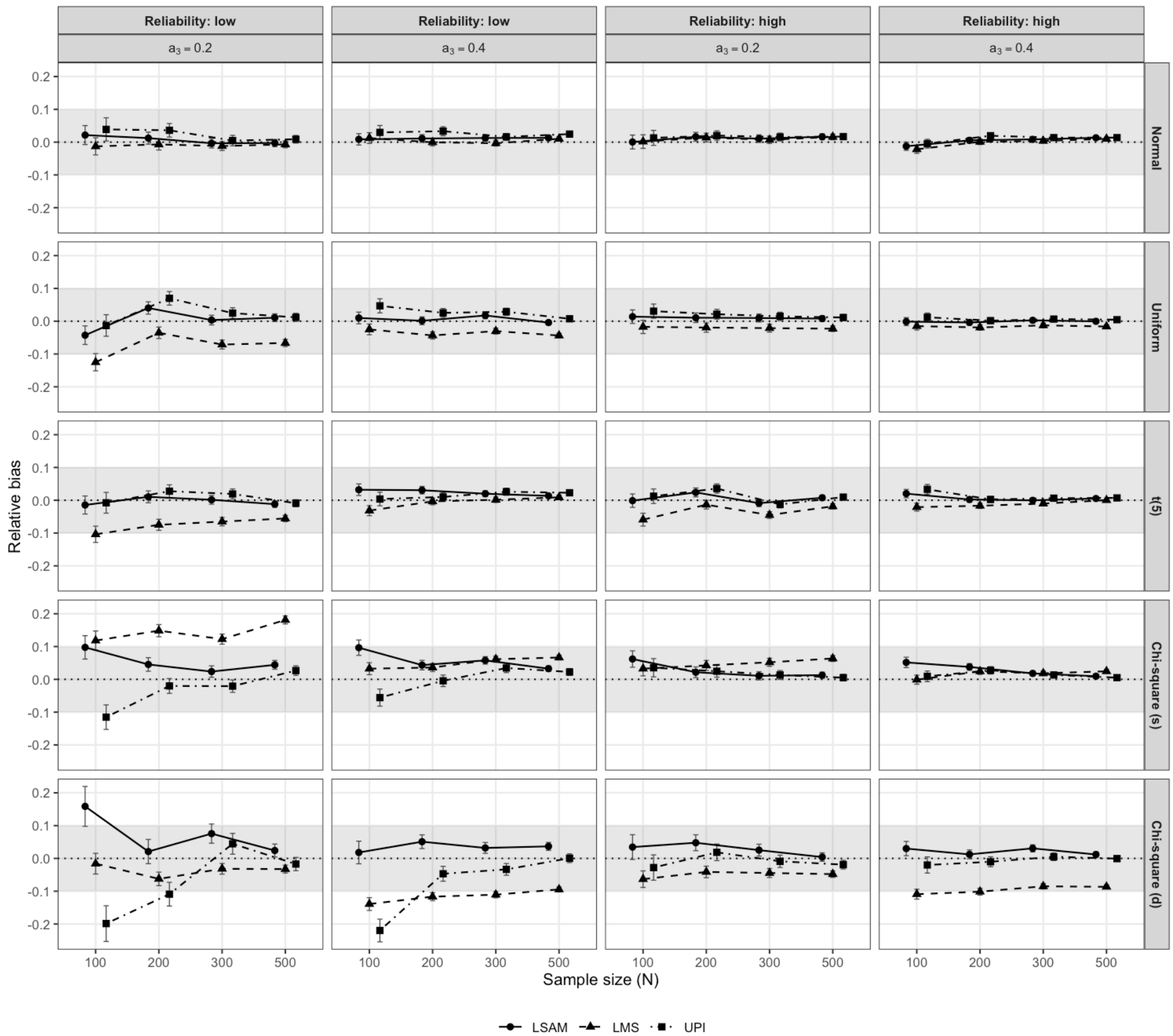
Population Model for the Simulation Study.



Note. X, W, M, and Y are latent variables, each measured by four congeneric indicators. Circles are latent variables and $X{\times}W$ is the latent interaction. Black elements form the correctly specified model that every estimator fits (a_1, a_2, a_3, b, c', and the X–W covariance). Grey elements are the misspecifications present in the population but omitted from the fitted model: two in the structural part, an omitted moderator effect on the outcome (c_2) and an omitted moderated direct effect (c_3); and two in the measurement part, an omitted cross-loading of X on m_3 (cl) and an omitted residual covariance between m_3 and y_3 (rc). Residual variances and intercepts are not shown.

Figure 2

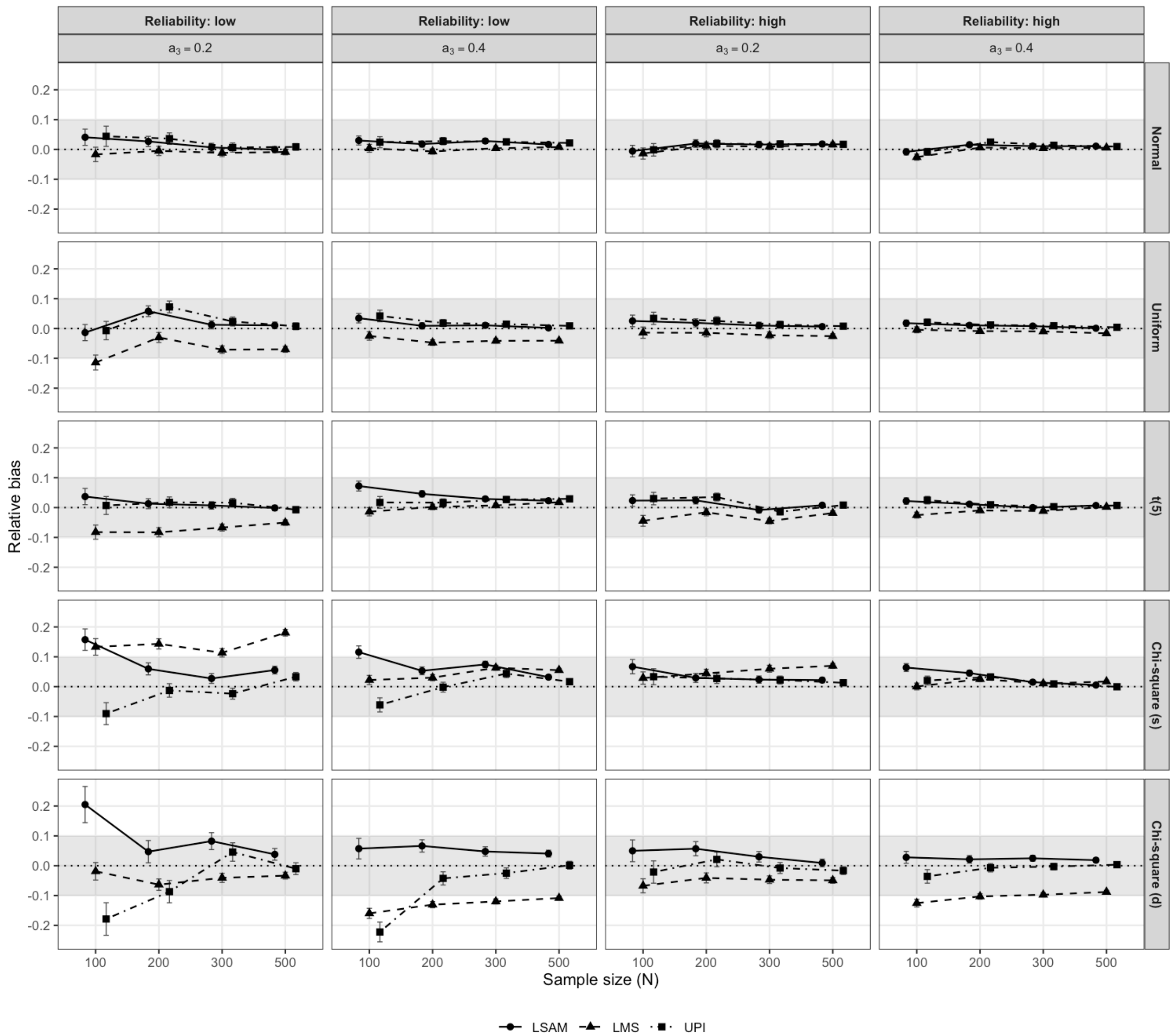
Relative Bias of the imm (a_3b) Under the Correctly Specified Condition Across Sample Sizes, Reliability, Interaction Values, and Distributions



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 3

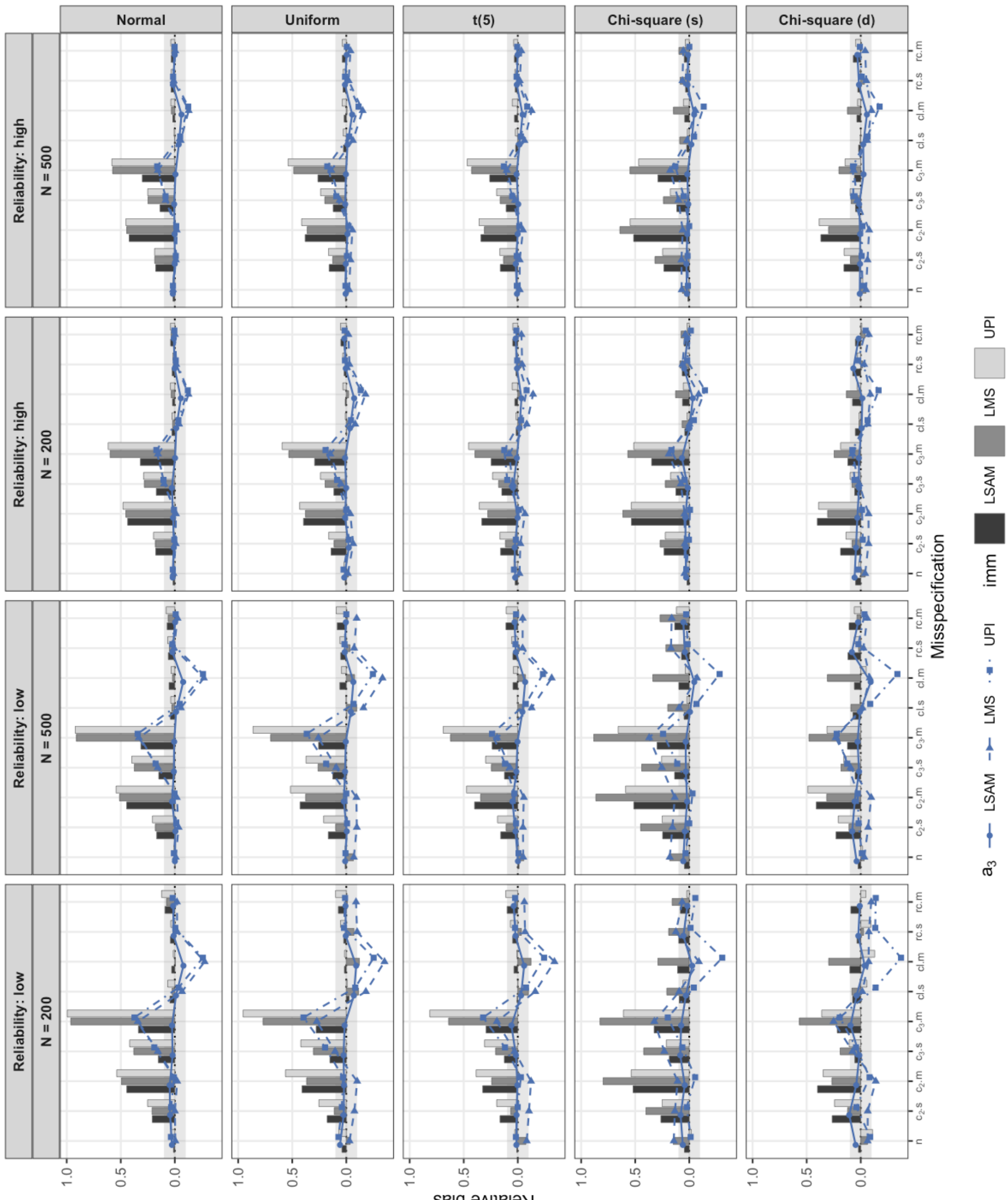
Relative Bias of the Interaction Coefficient (a_3) Under the Correctly Specified Condition Across Sample Sizes, Reliability, Interaction Values, and Distributions



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 4

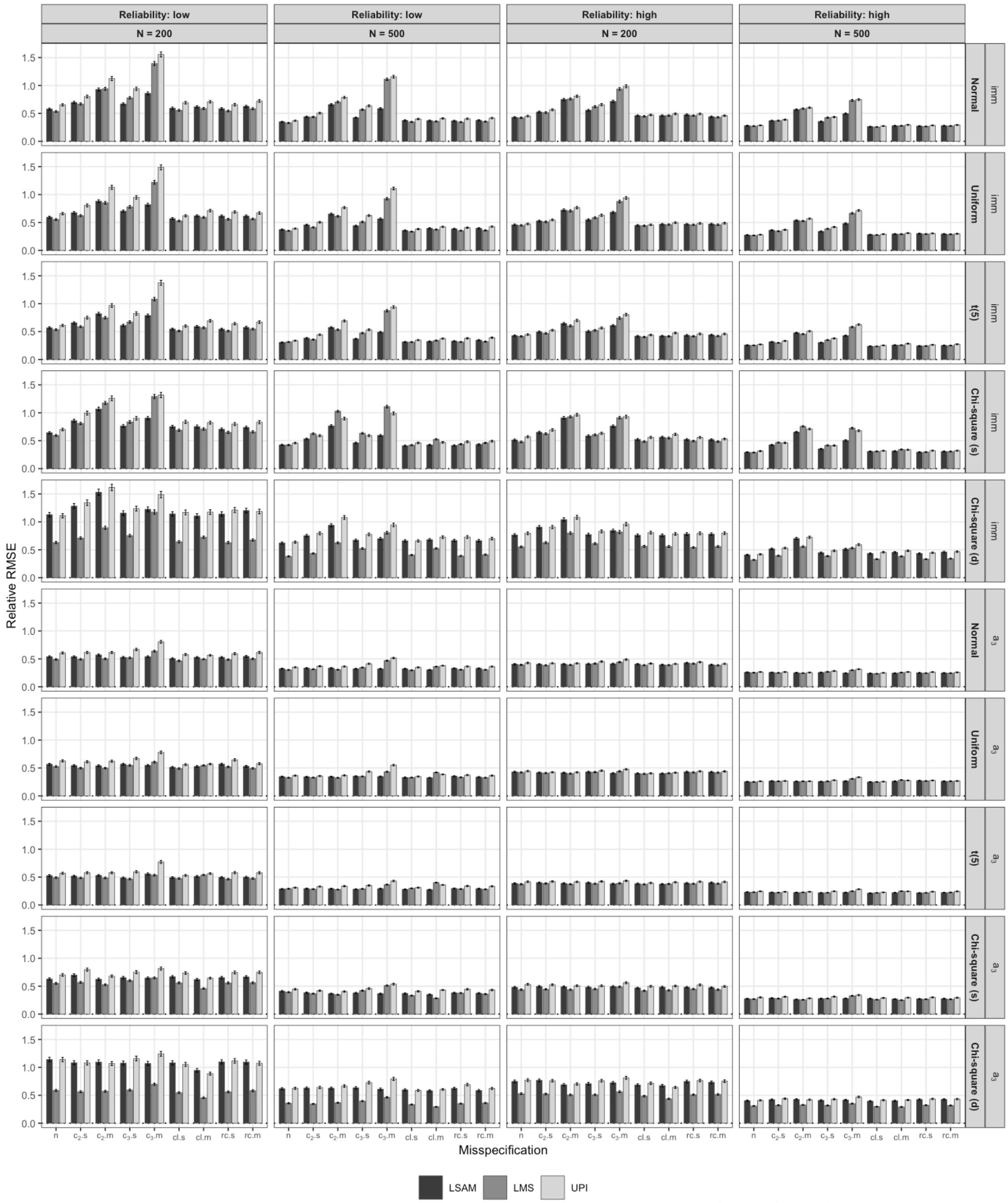
Relative Bias of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for relative bias. MCSEs are not shown.

Figure 5

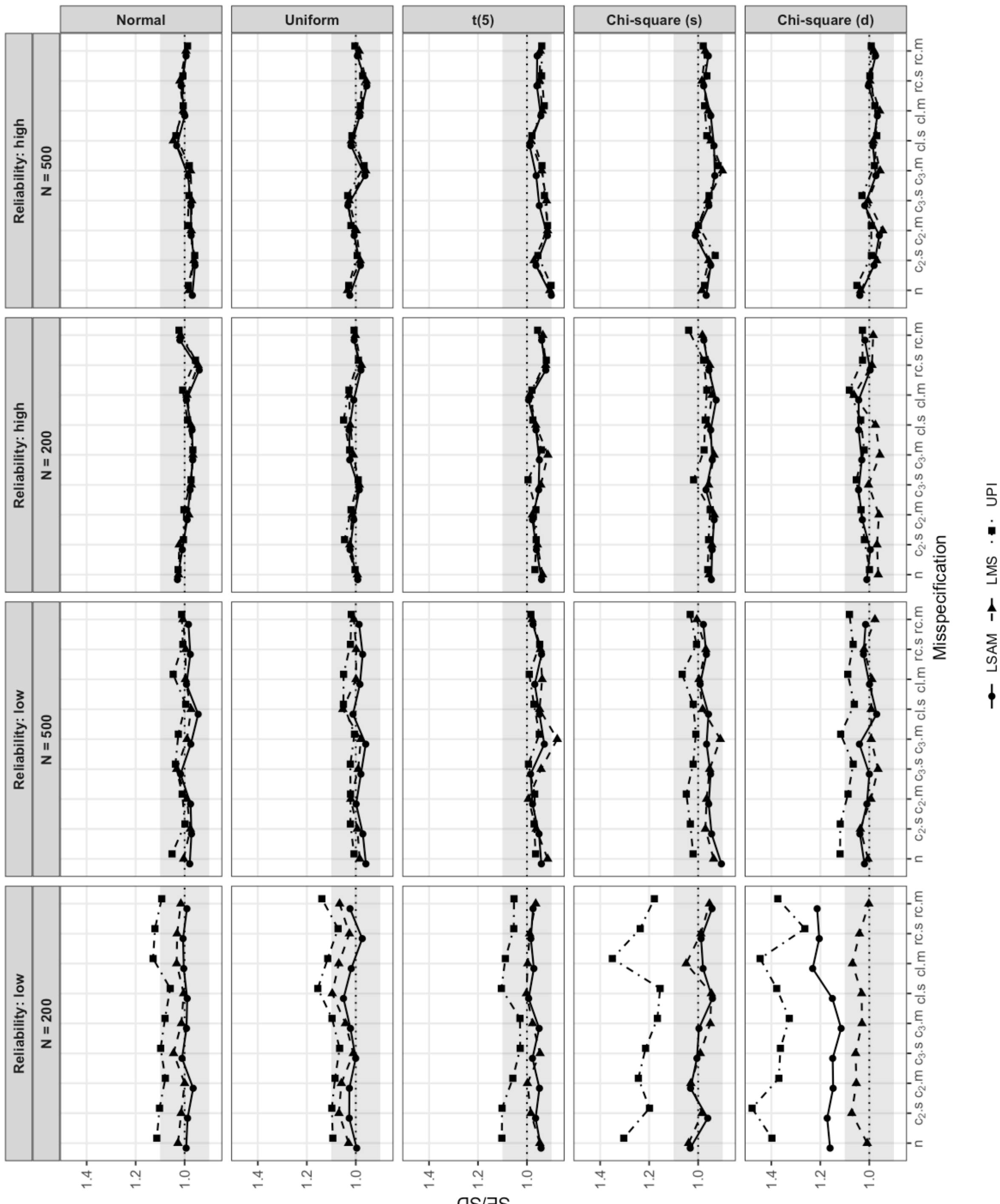
Relative RMSE of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The error bars represent the MCSE. No acceptable region is defined for this measure.

Figure 6

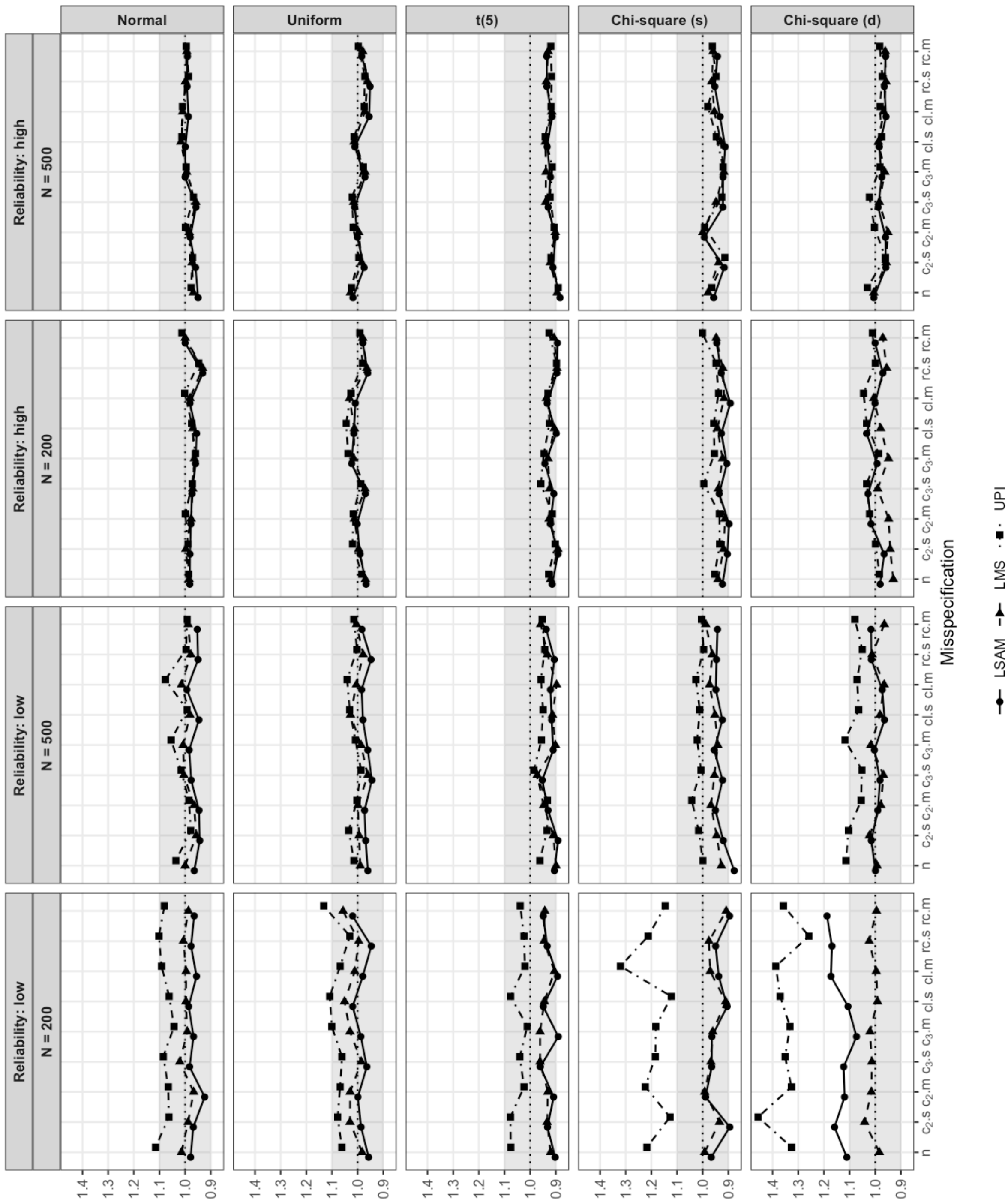
SE/SD of the imm (a_3b) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The grey band marks the acceptable region for the measure. The MCSE is not computed for the SE/SD ratio.

Figure 7

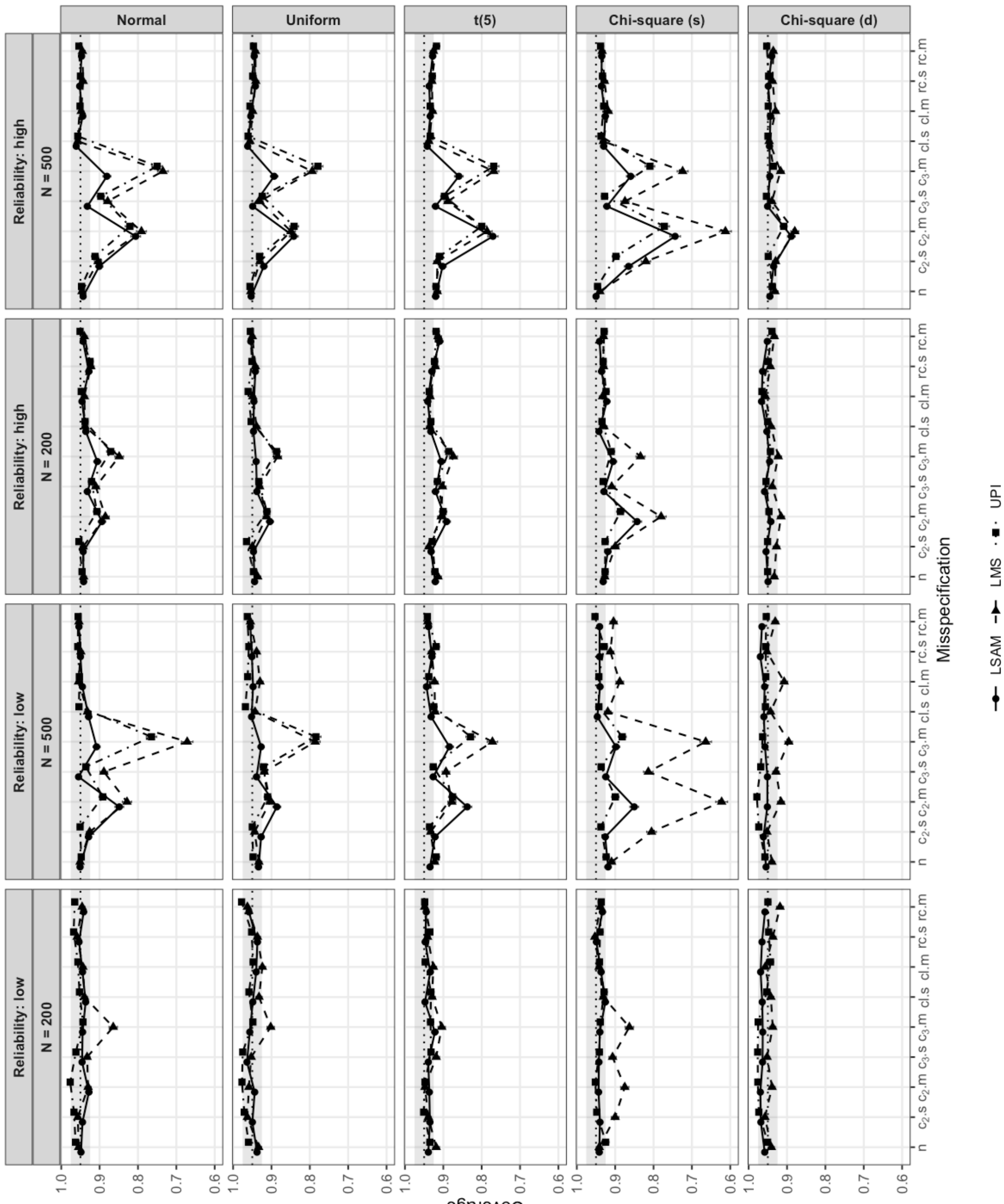
SE/SD of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for the measure. The MCSE is not computed for the SE/SD ratio.

Figure 8

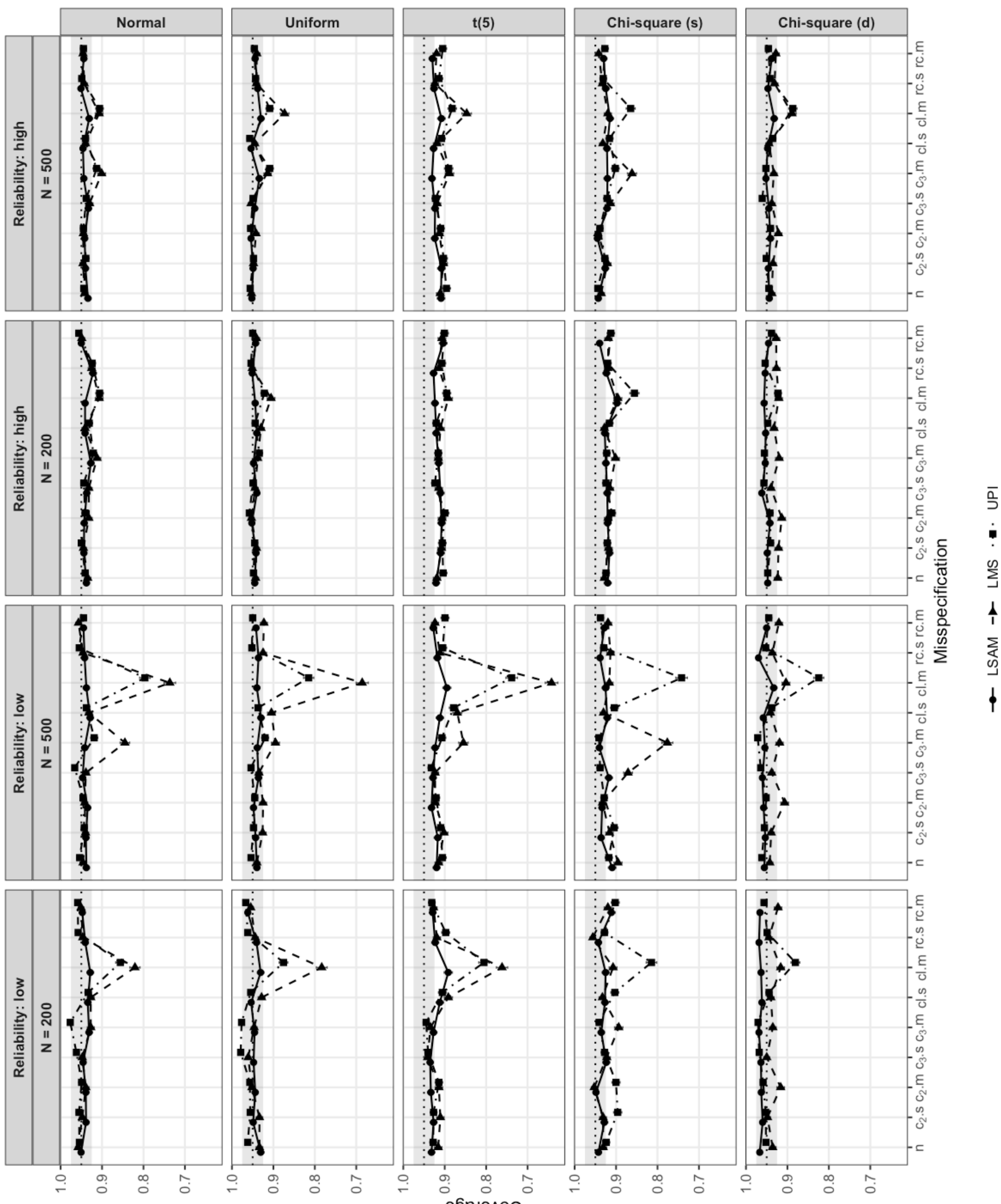
Coverage of the imm (a_3b) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 9

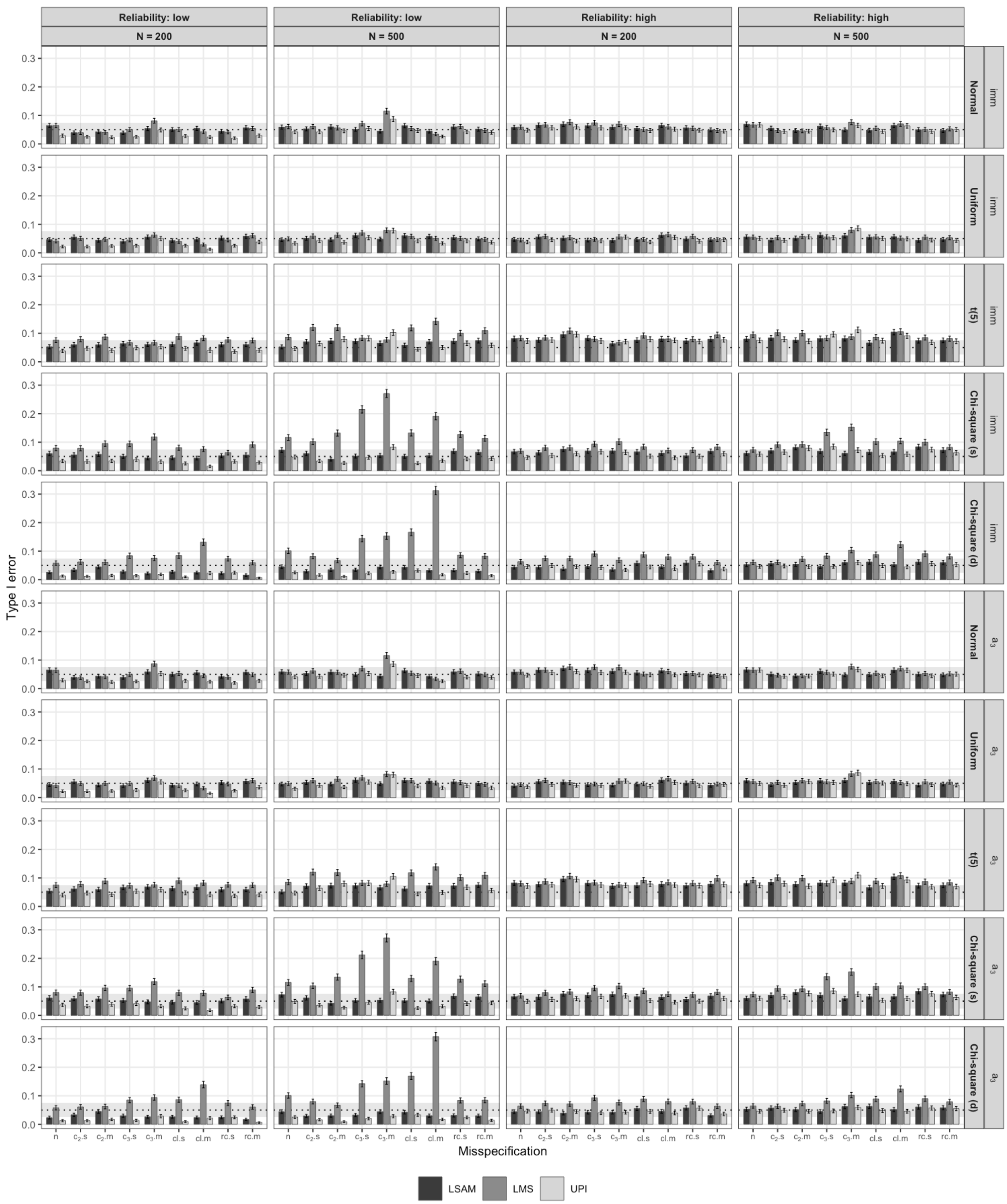
Coverage of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 10

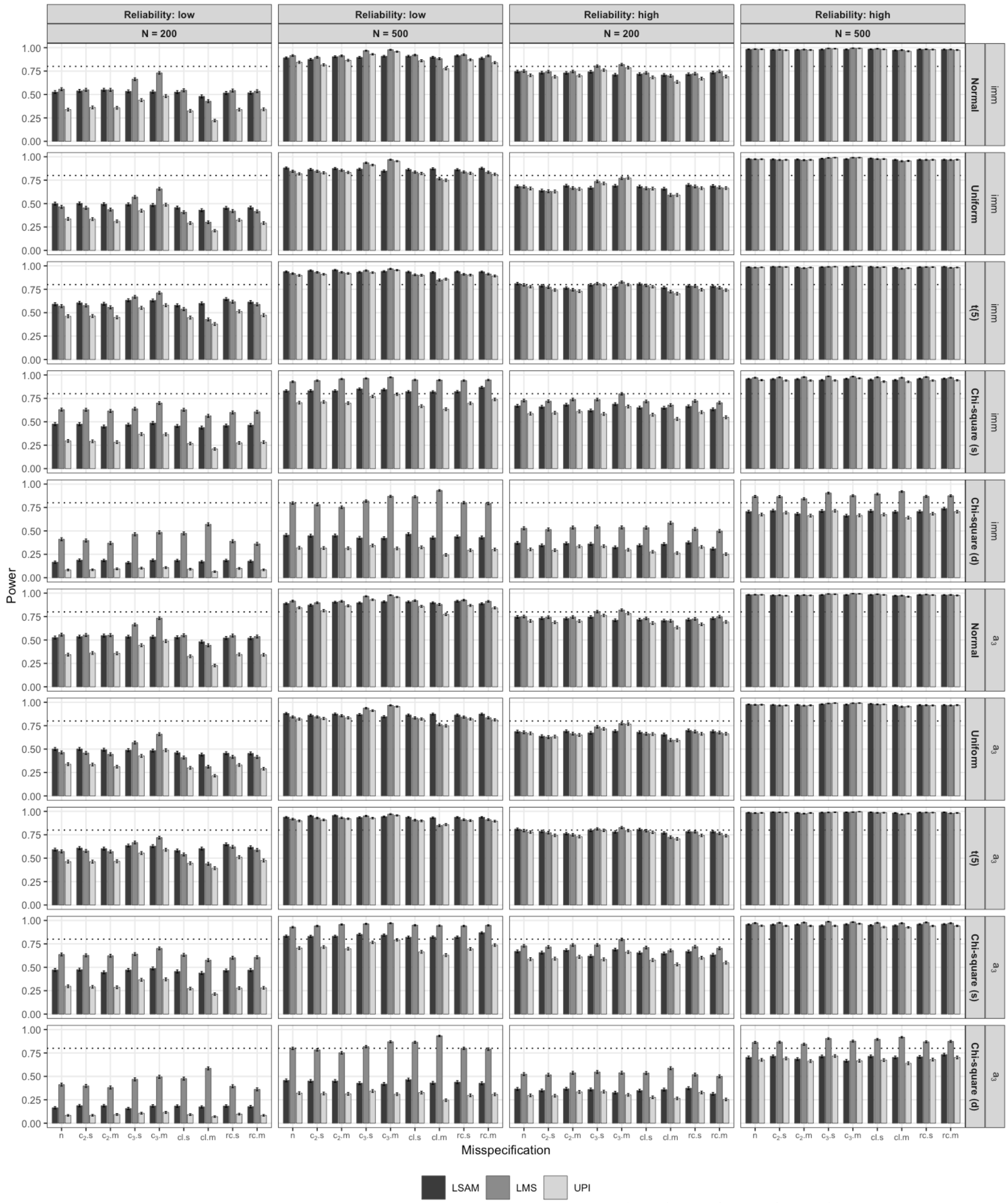
Type I Error of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0$



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 11

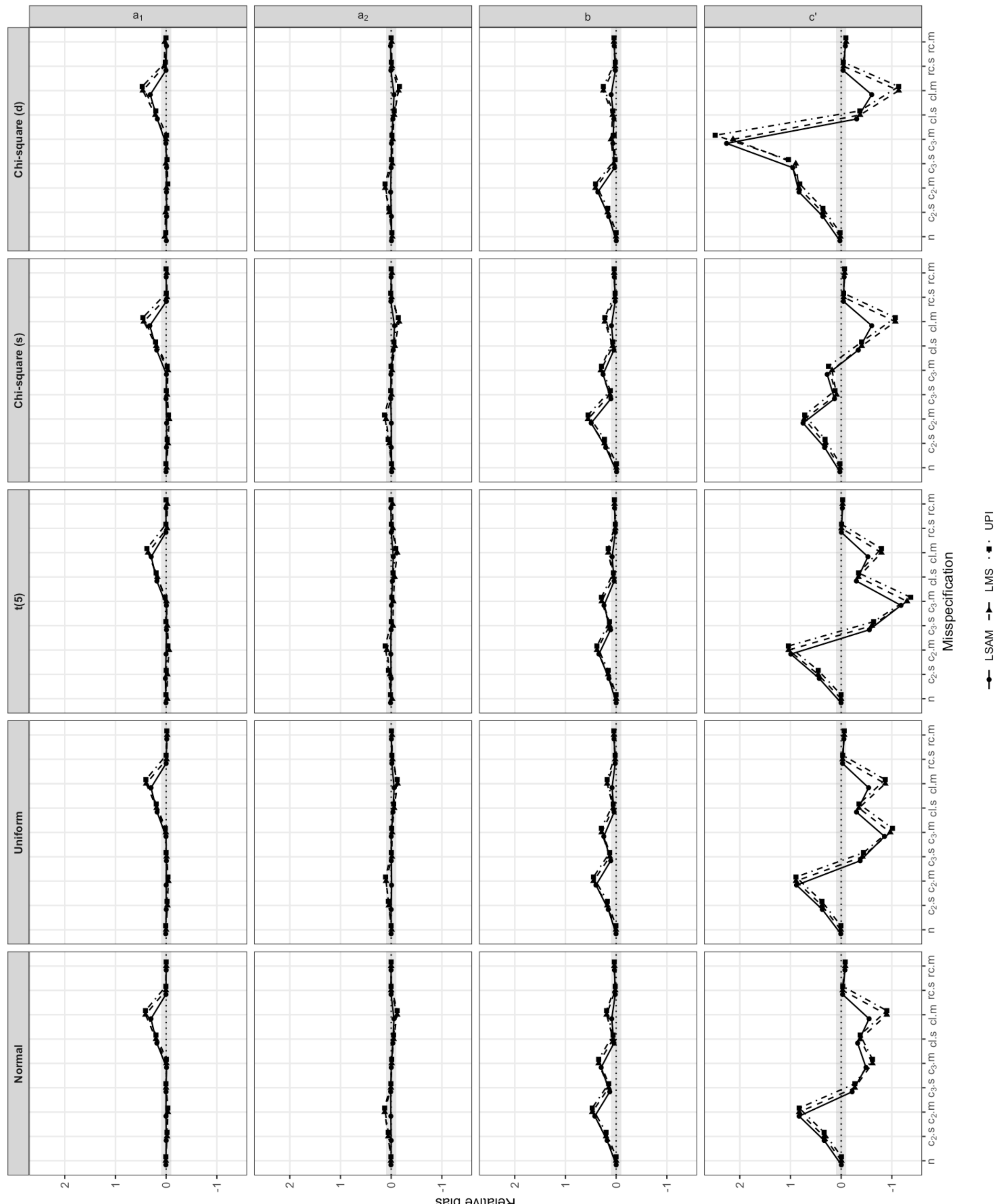
Power for the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The error bars represent the MCSE. No acceptable region is defined for this measure.

Figure 12

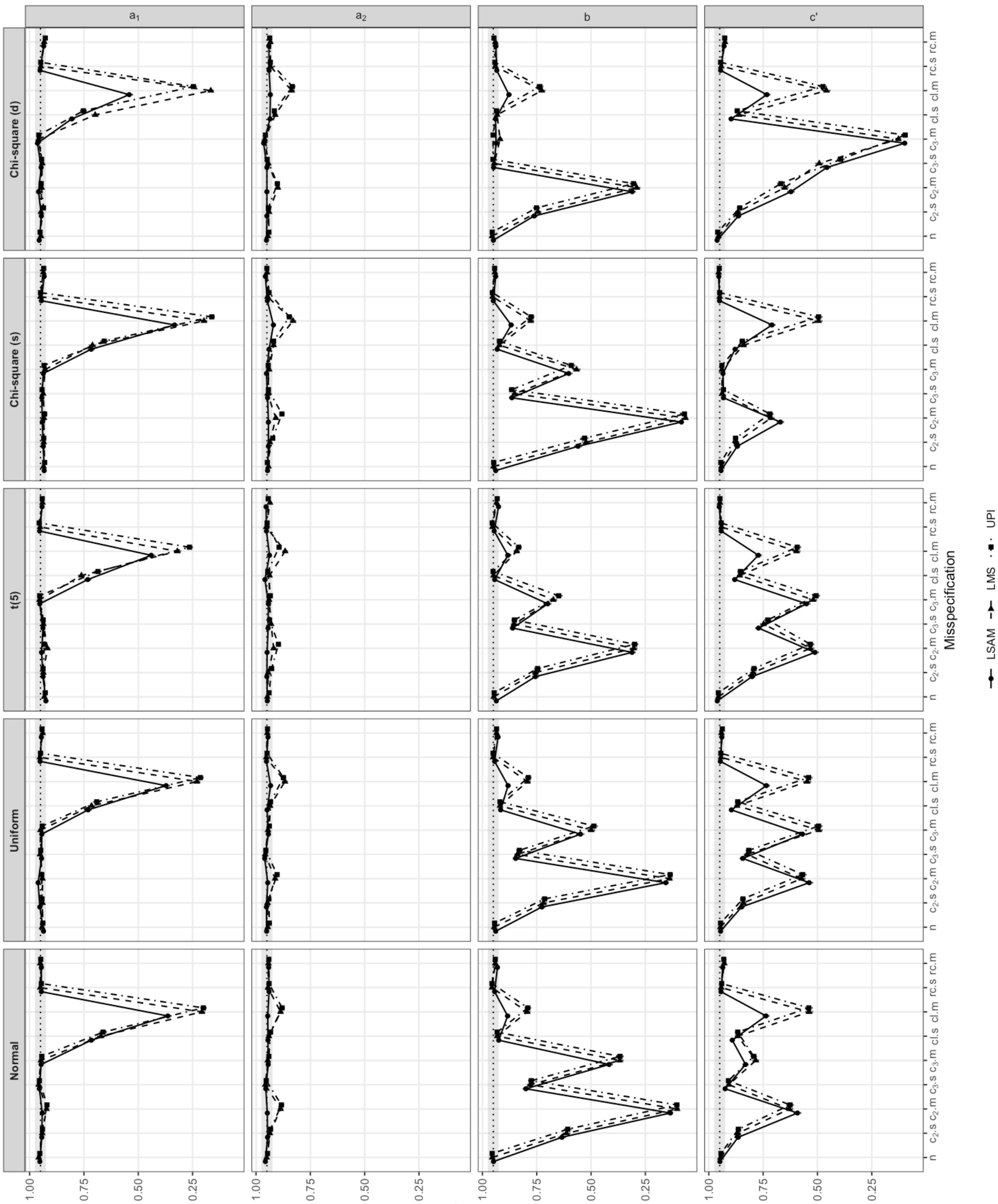
Relative Bias of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, N = 500, and a_3 = 0.2



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 13

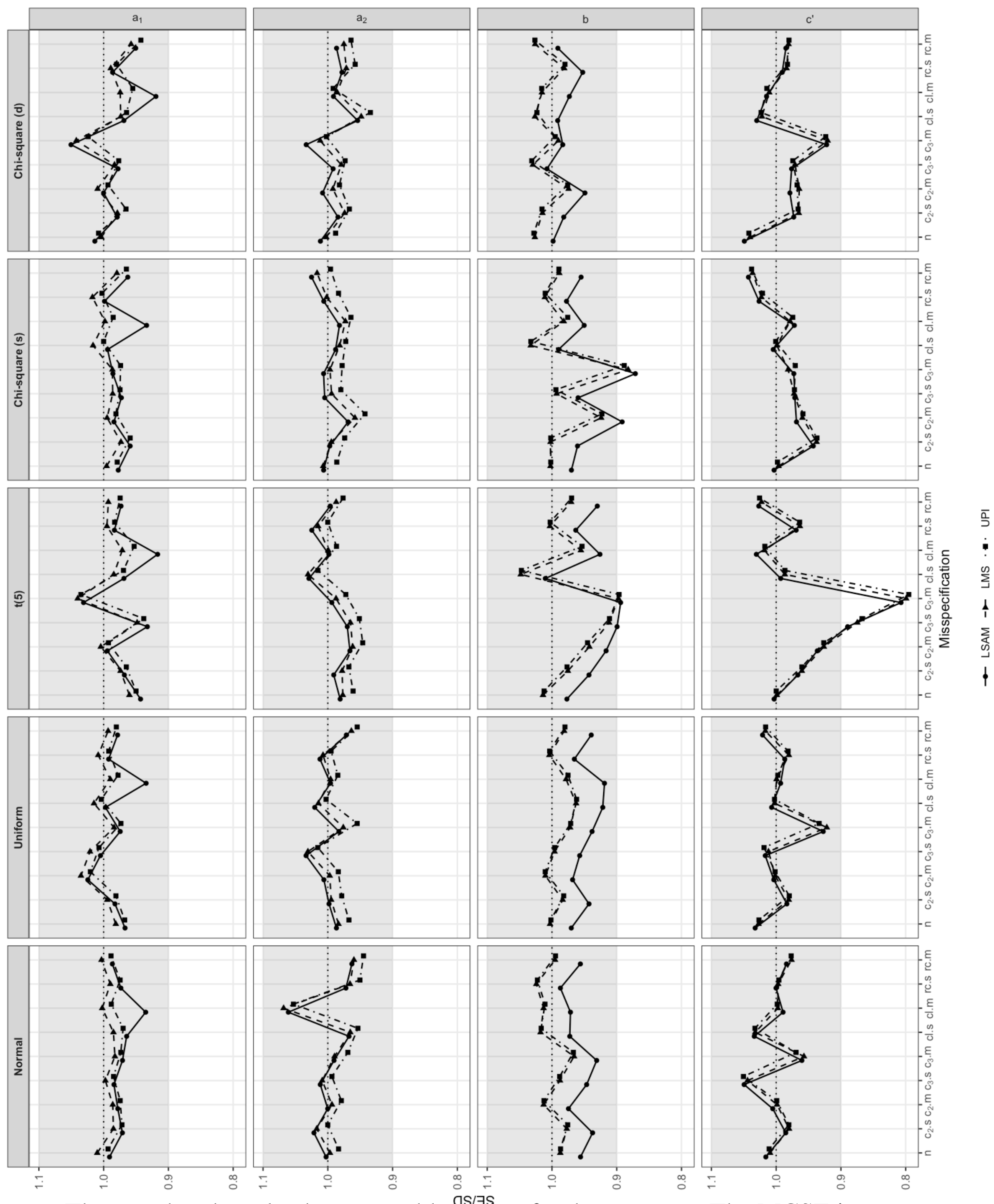
Coverage of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, N = 500, and a_3 = 0.2



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE.

Figure 14

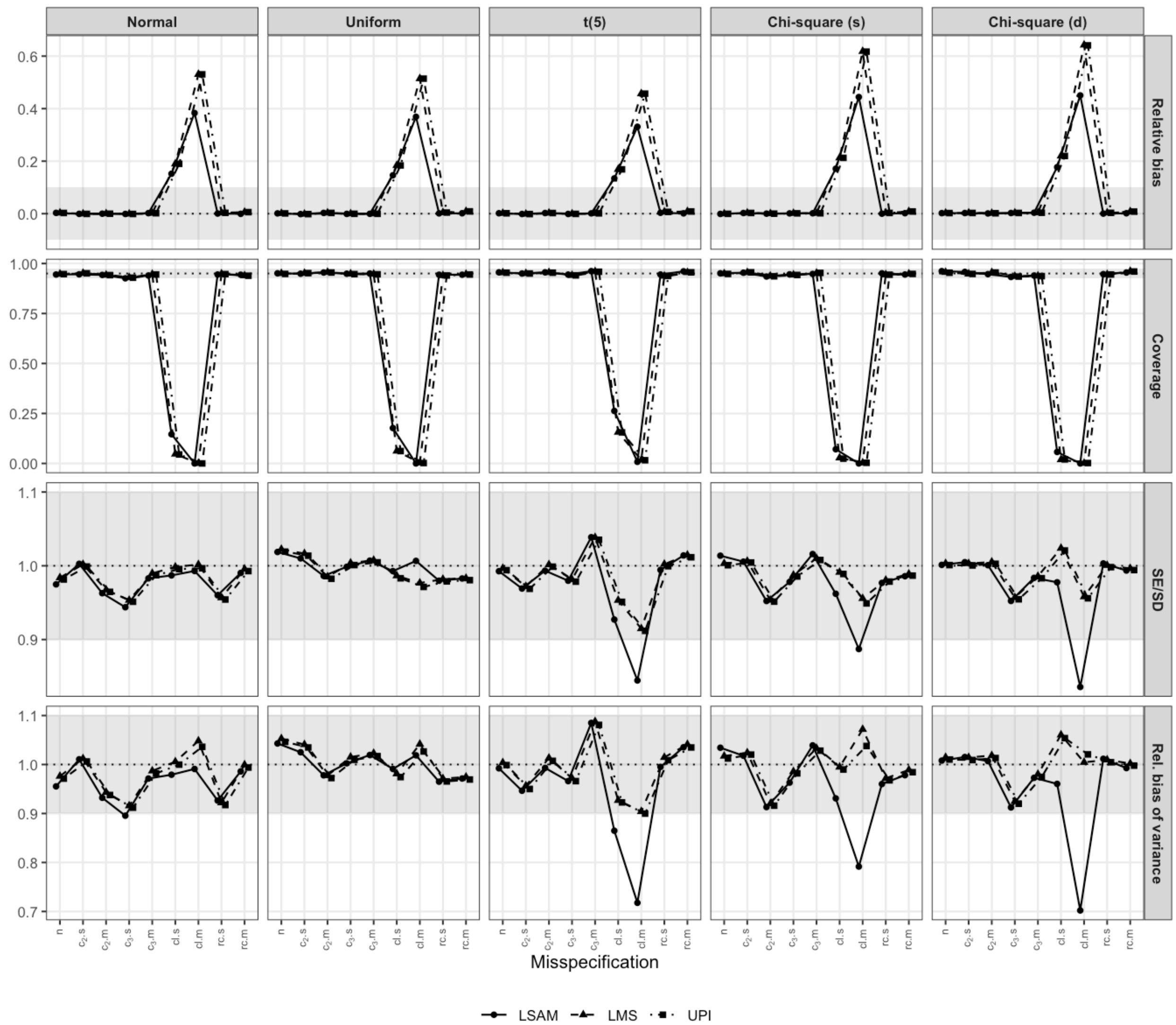
SE/SD of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, N = 500, and a_3 = 0.2



Note. The grey band marks the acceptable region for the measure. The MCSE is not computed for the SE/SD ratio.

Figure 15

Performance of the Cross-Loaded Mediator Indicator Loading (m_3) Across All Measures, Distributions, and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$



Note. The grey band in each panel marks the acceptable region for that measure.

Appendix A

Calibration and Data Generation

This appendix gives the formulas behind the data-generating mechanism. The full derivations, with worked numerical examples and even more details, are in the project repository. The calibration that fixes the variance parameters is given first, followed by the three-stage data generation that uses them.

Calibration

The structural disturbance variances and indicator error variances were fixed by the following closed-form calibration, computed once under the normal baseline.

Structural residual variances

The disturbance variances ψ_M, ψ_Y are fixed so that each structural equation explains a fraction $R^2 = 0.30$ of its outcome's variance at the central interaction level $a_3 = 0.2$ under the normal baseline. Writing the systematic part of M as $S_M = a_1 X + a_2 W + a_3(XW)$, its variance needs only second moments,

$$\mathrm{Var}(S_M) = a_1^2 + a_2^2 + 2a_1 a_2 \rho + a_3^2(1 + \rho^2),$$

and the disturbance is the remainder that brings the signal share to R^2,

$$\psi_M = \mathrm{Var}(S_M)\,\frac{1 - R^2}{R^2}, \qquad \psi_Y = \mathrm{Var}(bM + c'X)\,\frac{1 - R^2}{R^2},$$

with $\mathrm{Var}(bM + c'X) = b^2\,\mathrm{Var}(M) + c'^2 + 2bc'\,(a_1 + a_2\rho)$. Fixed at these values, the population variances of M and Y are closed-form functions of the parameters,

$$\mathrm{Var}(M) = \mathrm{Var}(S_M) + \psi_M, \qquad \mathrm{Var}(Y) = b^2\,\mathrm{Var}(M) + c'^2 + 2bc'\,(a_1 + a_2\rho) + \psi_Y,$$

evaluated once under the normal baseline and held fixed across distributions, so that R^2 and reliability drift only through the design factors.

Indicator reliabilities

To hit a target reliability $\omega \in \{0.5, 0.7\}$, the error variance of indicator j is set from the latent variance $V = \mathrm{Var}(\eta)$,

$$\theta_{\eta j} = \lambda_j^2 V \frac{1-\omega}{\omega}.$$

For X, W the variance is exactly 1; for M, Y it is the closed form above, fixed at the normal baseline, so the realized reliability drifts mildly across distributions.

Calibrating the misspecification magnitudes

The *small* and *medium* labels are standardized coefficients, $t \in \{0.25, 0.50\}$. For an omitted predictor p added to a baseline outcome y_0, the magnitude is the fully standardized regression coefficient of p,

$$t = \frac{\beta \ \mathrm{sd}(p)}{\mathrm{sd}(y_0 + \beta p)}.$$

Squaring and substituting $\mathrm{Var}(y_0 + \beta p) = \mathrm{Var}(y_0) + \beta^2 \mathrm{Var}(p) + 2\beta \mathrm{Cov}(p, y_0)$ yields a quadratic $A\beta^2 + B\beta + C = 0$ with

$$A = \mathrm{Var}(p)\,(1-t^2), \quad B = -2t^2\,\mathrm{Cov}(p, y_0), \quad C = -t^2\,\mathrm{Var}(y_0).$$

Since $A > 0$ and $C < 0$ the roots have opposite sign; the positive omitted path is the positive root,

$$\beta = \frac{t^2\,\mathrm{Cov}(p, y_0) + \sqrt{t^4\,\mathrm{Cov}(p, y_0)^2 + t^2(1-t^2)\ \mathrm{Var}(p)\ \mathrm{Var}(y_0)}}{\mathrm{Var}(p)\,(1-t^2)}.$$

The three moments are estimated from one large ($n = 10^6$) sample per distribution and a_3 level. The predictor p and baseline y_0 differ by misspecification,

$$
\begin{aligned}
c_2: &\quad p = W, &\quad y_0 &= bM + c'X + \varepsilon_Y,\\
c_3: &\quad p = XW, &\quad y_0 &= bM + c'X + \varepsilon_Y,\\
\texttt{cl}: &\quad p = X, &\quad y_0 &= m_3 = \lambda_3 M + \delta_{m_3}.
\end{aligned}
$$

c_2 and c_3 are reliability independent; `cl` depends on ω because its baseline m_3 carries measurement noise. `rc` never enters the calibration: it is a residual correlation and equals its target in every cell.

Holding Y reliability fixed under c_2 and c_3

Adding an omitted structural path inflates $\mathrm{Var}(Y)$ by $\kappa = \mathrm{Var}(y_0 + \beta p)/\mathrm{Var}(y_0)$, which would raise Y's indicator reliability as a side effect. Scaling the Y-indicator noise by the same κ makes signal and noise move together, so κ cancels and the reliability returns to its no-misspecification value,

$$
\omega_j = \frac{\kappa \lambda_j^2 V}{\kappa \lambda_j^2 V + \kappa \theta_j} = \frac{\lambda_j^2 V}{\lambda_j^2 V + \theta_j}.
$$

The latent effect size is untouched; only the Y indicators are made correspondingly noisier. The measurement level misspecifications `cl` and `rc` act on the indicators directly and are not compensated, since that contamination is itself the misspecification.

Data generation

Using the calibrated variances above, data were generated in three stages: the correlated exogenous predictors, the dependent latent variables through the structural model, and the observed indicators through the measurement model.

Stage I: Exogenous predictors

The correlated predictors (X, W) were generated with the vine-to-anything (VITA) method (Foldnes & Grønneberg, 2015; Grønneberg et al., 2022), as implemented in the *covsim* package, with sampling via *rvinecopulib* (Nagler & Vatter, 2025). They were specified to have mean 0, variance 1, and the covariance structure

$$\Phi = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.3 \\ 0.3 & 1 \end{bmatrix},$$

so that the conditions differ only in marginal shape. Under the normal condition, the predictors had Gaussian marginals with the specified covariance structure. Under the uniform condition, they had symmetric uniform marginals $\mathrm{U}(-\sqrt{3}, \sqrt{3})$, which have mean 0 and variance 1. Under the t_5 condition, they had Student t marginals with five degrees of freedom, rescaled by $\sqrt{5/3}$ to unit variance. The two χ^2 conditions used χ^2_2 marginals, centered by their mean (2) and scaled by their standard deviation (2) to mean 0 and variance 1. In the same-direction condition both predictors are right-skewed, whereas in the opposite-direction condition W is negated so that X and W are skewed in opposite directions, with the sampled correlation sign flipped to restore $\mathrm{Cor}(X, W) = \rho$. In every condition, VITA constructed a vine copula to impose Φ while preserving the marginals.

Stage II: Structural model

The dependent latent variables were generated from the structural equations

$$M = a_1 X + a_2 W + a_3 (X \times W) + \zeta_M, \qquad Y = bM + c'X + \zeta_Y,$$

with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and $\zeta_Y \sim \mathcal{N}(0, \psi_Y)$. The disturbance variances ψ_M, ψ_Y take the calibrated values so that each equation explains $R^2 \approx 0.30$ of its outcome's variance at the central interaction level under the normal baseline.

Stage III: Measurement model

Each latent $\eta \in \{X, W, M, Y\}$ was measured by four indicators,

$$z_{\eta j} = \lambda_j \eta + \delta_{\eta j}, \qquad \lambda = (1, 0.8, 0.7, 0.6),$$

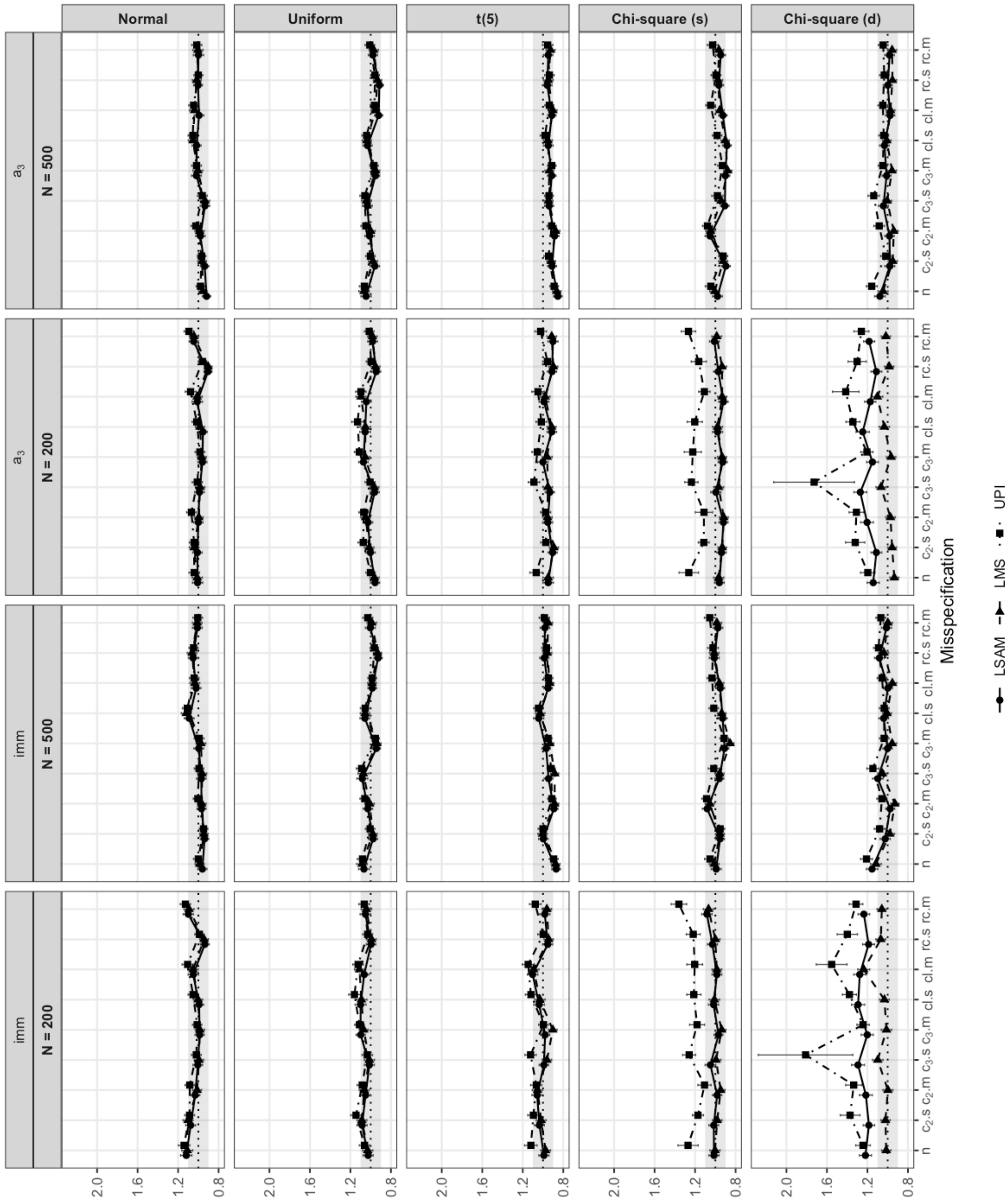
with independent Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$. The error variances $\theta_{\eta j}$ take the calibrated values that achieve the target reliability $\omega \in \{0.5, 0.7\}$. Only the exogenous predictors varied in distribution. The structural disturbances and measurement errors were Gaussian in every condition.

Appendix B

Relative Bias of the Variance Estimator at Low Reliability

Figure B1

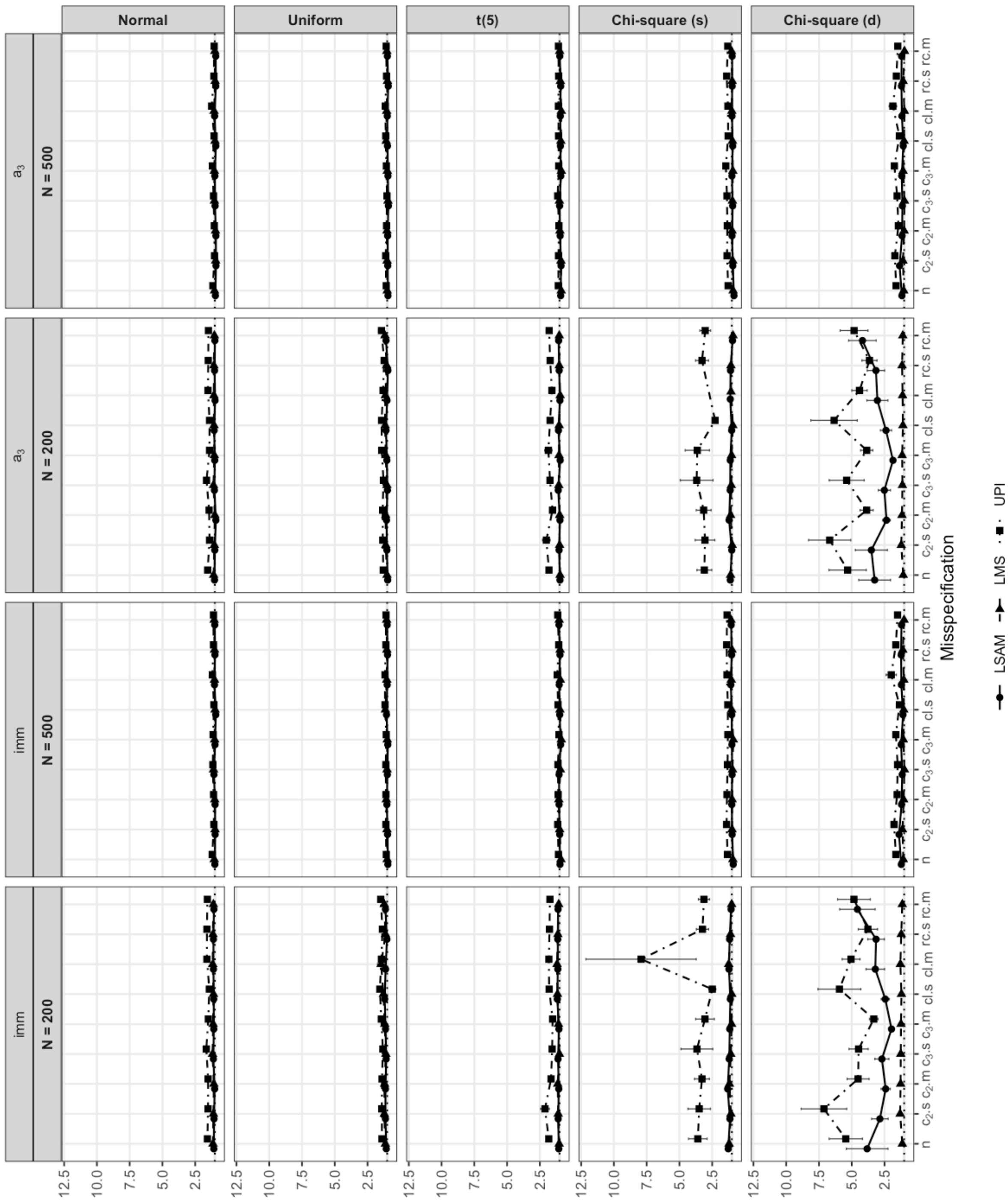
Relative Bias of the Variance Estimator of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions and (Mis)specifications for $N = 200$ and 500 at High Reliability and $a_3 = 0.2$



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE. Cells above ten are truncated.

Figure B2

Relative Bias of the Variance Estimator of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions and (Mis)specifications for N = 200 and 500 at Low Reliability and a_3 = 0.2



Note. The grey band marks the acceptable region for the measure, and the error bars represent the MCSE. Cells above ten are truncated.